This is the free Material Data Center Datasheet of Ultramid® B3S - PA6 - BASF

Product Texts

An easy flowing, finely crystalline injection moulding grade for very fast processing. Parts produced include thin-walled technical parts (eg housing, fittings, grips, small parts and fixing clamps).

Processing/Physical Characteristics	dry / cond	Unit	Test Standard
ISO Data			
Melt volume-flow rate, MVR	160 / *	cm³/10min	ISO 1133
Temperature	275 / *	°C	-
Load	5 / *	kg	-
Molding shrinkage, parallel	0.9 / *	%	ISO 294-4, 2577
Molding shrinkage, normal	0.9 / *	%	ISO 294-4, 2577
Ejection temperature	145	°C	-

Mechanical properties	dry / cond	Unit	Test Standard
ISO Data			
Tensile Modulus	3500 / 1200	MPa	ISO 527
Yield stress	90 / 45	MPa	ISO 527
Yield strain	4 / 20	%	ISO 527
Nominal strain at break	10 / >50	%	ISO 527
Tensile creep modulus, 1h	* / 1300	MPa	ISO 899-1
Tensile creep modulus, 1000h	* / 1100	MPa	ISO 899-1
Charpy impact strength, +23°C	250 / N	kJ/m²	ISO 179/1eU
Charpy impact strength, -30°C	200 / -	kJ/m²	ISO 179/1eU
Charpy notched impact strength, +23°C	4 / 50	kJ/m²	ISO 179/1eA
Charpy notched impact strength, -30°C	3 / -	kJ/m²	ISO 179/1eA

Thermal properties	dry / cond	Unit	Test Standard
ISO Data			
Melting temperature, 10°C/min	220 / *	°C	ISO 11357-1/-3
Glass transition temperature, 10°C/min	60 / *	°C	ISO 11357-1/-2
Temp. of deflection under load, 1.80 MPa	65 / *	°C	ISO 75-1/-2
Temp. of deflection under load, 0.45 MPa	180 / *	°C	ISO 75-1/-2
Vicat softening temperature, B	204 / *	°C	ISO 306
Coeff. of linear therm. expansion, parallel	102 / *	E-6/K	ISO 11359-1/-2
Oxygen index	26 / *	%	ISO 4589-1/-2

Electrical properties	dry / cond	Unit	Test Standard
ISO Data			
Relative permittivity, 100Hz	4.1 / -	-	IEC 62631-2-1
Relative permittivity, 1MHz	3.3 / 7	-	IEC 62631-2-1
Dissipation factor, 100Hz	100 / -	E-4	IEC 62631-2-1
Dissipation factor, 1MHz	300 / 3000	E-4	IEC 62631-2-1
Volume resistivity	1E13 / 1E10	Ohm*m	IEC 62631-3-1
Surface resistivity	* / 1E10	Ohm	IEC 62631-3-2
Electric strength	40 / 29	kV/mm	IEC 60243-1
Comparative tracking index	- / 600	-	IEC 60112

Other properties	dry / cond	Unit	Test Standard
Water absorption	9.5 / *	%	Sim. to ISO 62
Humidity absorption	3 / *	%	Sim. to ISO 62
Density	1130 / -	kg/m³	ISO 1183

Material specific properties	dry / cond	Unit	Test Standard
ISO Data			
Viscosity number	145 / *	cm³/g	ISO 307, 1157, 1628

Test specimen production	Value	Unit	Test Standard
ISO Data			
Injection Molding, melt temperature	250	°C	ISO 294
Injection Molding, mold temperature	80	°C	ISO 294
Injection Molding, injection velocity	200	mm/s	ISO 294

Processing Recommendation Injection Molding	Value	Unit	Test Standard
Pre-drying - Temperature	80	°C	-
Pre-drying - Time	4	h	-
Processing humidity	≤0.15	%	-
Melt temperature	250 - 270	°C	-
Mold temperature	40 - 60	°C	-

Characteristics

Processing
Injection Molding

Delivery form
Pellets

Additives
Lubricants, Release agent

Regional Availability
North America, Europe, Asia Pacific, South and Central America, Near East/Africa

Other text information

Injection molding

PREPROCESSING
Pre/Post-processing, max. allowed water content: .15 %
Pre/Post-processing, Pre-drying, Temperature: 80 °C
Pre/Post-processing, Pre-drying, Time: 4 h

PROCESSING
injection molding, Melt temperature, range: 250 - 270 °C
injection molding, Melt temperature, recommended: 260 °C
injection molding, Mold temperature, range: 40 - 60 °C
injection molding, Mold temperature, recommended: 60 °C
injection molding, Dwell time, thermoplastics: 10 min

Application examples

Coatrack
Handle, door
Screwless electric terminals

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